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Research Article submission

Dear Editors of IEEE Access,

Please consider our manuscript, “Protocol-Constrained Residual Adaptation for Rigid-Body Inference: A Bounded Evaluation of Frozen Visual Features,” by Hyojin Park, Nam-Hyun Yoo, and Jinhong Yang, for publication as a **Research Article**.

The study examines whether a compact visual residual can add predictive information after a strong numerical estimator has already used a short motion observation. The method combines fixed robust estimation, numerical residual adaptation, and a bounded frozen-feature residual with protocol-specific permissions. Its contribution is the tested allocation of roles and the measured increment, rather than a claim that hybrid estimation or VLM-based physical inference is new in isolation.

On 72 new synthetic geometry families, the complete method reduces analytic trajectory-projection error by 29.95% relative to the one-step-weight Huber anchor. A post hoc iterated-Huber comparison retains 24.11% improvement. Most gain comes from sliding numerical adaptation. The original visual increment is 1.07%; cross-fitted targets retain 1.17%, with both two-sided intervals including zero. Development-selected channel controls associate the increment primarily with decoder position corrections, without isolating language pretraining.

A controlled PyBullet sphere proxy retains a 28.19% reduction and high agreement with analytic family rankings. The paper reports direct-regression baselines, protocol/coordinate breakdowns, covariance sensitivity, and measured latency components. These added comparisons are explicitly post hoc; they do not replace the original models or confirmation rules.

A public pushing task yields a 4.27% visual-plus-gate increment. Separating both object and surface friction conditions during refitting retains 4.38% material-pair improvement on 329 episodes, with interval $[-6.044, -0.116]$ mm; its upper endpoint is near zero. All episodes share cube/scene assets. The original 3.90% development result remains below its 10% criterion, and the tested real-video score fails directionally. These boundaries are stated alongside the positive results.

We believe that this structured estimation method and its inspectable evaluation are relevant to the engineering readership of IEEE Access. The initial-submission package includes manuscript source, supplementary methods, figures, and versioned experimental verification materials at GitHub release v0.3.0. Funding and AI assistance are disclosed in the manuscript.

Thank you for considering this work.

Sincerely,
Jinhong Yang
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